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Chemical remagnetization of the Upper Carboniferous–Lower Triassic Pyeongan Supergroup in the Jeongseon area, Korea: fluid migration through the Ogcheon Fold Belt

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SUMMARY

Palaeomagnetic and rock magnetic investigations have been carried out for the Upper Carboniferous–Lower Triassic Pyeongan Supergroup, exposed in the Jeongseon area in eastern South Korea. A total of 302 independently oriented core samples were drilled from 24 sites for the study. The mean direction in stratigraphic coordinates ($D/I = 345.3^\circ/38.4^\circ$, $k = 4.8$, $\alpha_{95} = 18.2^\circ$) is more dispersed than the mean direction in geographic coordinates ($D/I = 354.9^\circ/61.5^\circ$, $k = 58.0$, $\alpha_{95} = 4.7^\circ$), and the stepwise untilting of the characteristic remanent magnetization (ChRM) reveals a maximum value of k at 0 per cent untilting. Furthermore, authigenic magnetic mineral grains accompanied with altered clays are identified by electron microscope observations. These results collectively imply that the ChRMs of this study were acquired by the formation of authigenic magnetic minerals after tilting of the strata. The palaeomagnetic pole position (97.3°E, 82.4°N, $A_{95} = 6.8^\circ$) of the Pyeongan Supergroup calculated from the site mean directions of the ChRMs in geographic coordinates is close to those of the Late Cretaceous and Tertiary periods of the Korean Peninsula. The presence of many hydrothermal vein deposits, reportedly formed between 94 and 73 Ma, near the study area and the evidence for extensive alteration of clay minerals under the influence of fluids shown in the X-ray diffraction study strongly suggest that the chemical remanent magnetization was acquired by a fluid-mediated process during the Late Cretaceous–Early Tertiary. Considering the different remagnetization timing of several areas in the Ogcheon Fold Belt, which extends from NE to SW diagonally across the Korean Peninsula, it is interpreted that the fluids and magmatism associated with this remagnetization originated from the subduction of the Kula/Pacific plates under the Eurasian Plate during the Cretaceous period. The fluids might have migrated northward or northeastward through the fault systems within the Ogcheon Fold Belt.

Key words: chemical remanent magnetization, fluid migration, Korea, palaeomagnetism, Pyeongan Supergroup, remagnetization.

1 INTRODUCTION

The Ogcheon Fold Belt, which is located between the Gyeonggi Massif to the north and the Ryeongnam Massif to the south (Fig. 1), has been regarded as one of the most important regions within the Korean Peninsula to many geologists on account of the ambiguity of its origin and stratigraphy. Also, whether the eastern extension of the Quingling–Dabie Suture Zone between the North China Block and the Yangtze Block (South China Block) is the Ogcheon Fold Belt (Oh *et al.* 1999) or the Imjingang Belt (Ree *et al.* 1999) is still controversial. A palaeomagnetic investigation of the Ogcheon Fold Belt might provide clues to these unsolved geological problems and tectonic history of East Asia and those of the Korean Peninsula. The Ogcheon Fold Belt is divided into two parts: the metamorphosed southwestern subzone and the unmetamorphosed northeastern subzone (Lee 1988). Over the previous decade, many palaeomagnetic studies of Palaeozoic and Mesozoic rocks in the northeastern subzone were carried out (Shibuya *et al.* 1988; Kim 1989; Otofuji *et al.* 1989; Kim & Van der Voo 1990; Kim *et al.* 1992; Min *et al.* 1993; Doh & Piper 1994; Lee *et al.* 1996, 1997; Doh *et al.* 1997, 1998; Uno 1999, 2000). Some results, however, are questionable because the number of samples was insufficient to represent the long geological time interval. More importantly, widespread Cretaceous to Tertiary remagnetizations were identified within the northeastern subzone (e.g. the Yeongwol area, Doh *et al.* 1997; the Pyeongchang area, Park *et al.* 2000) and in some Cretaceous basins lying along

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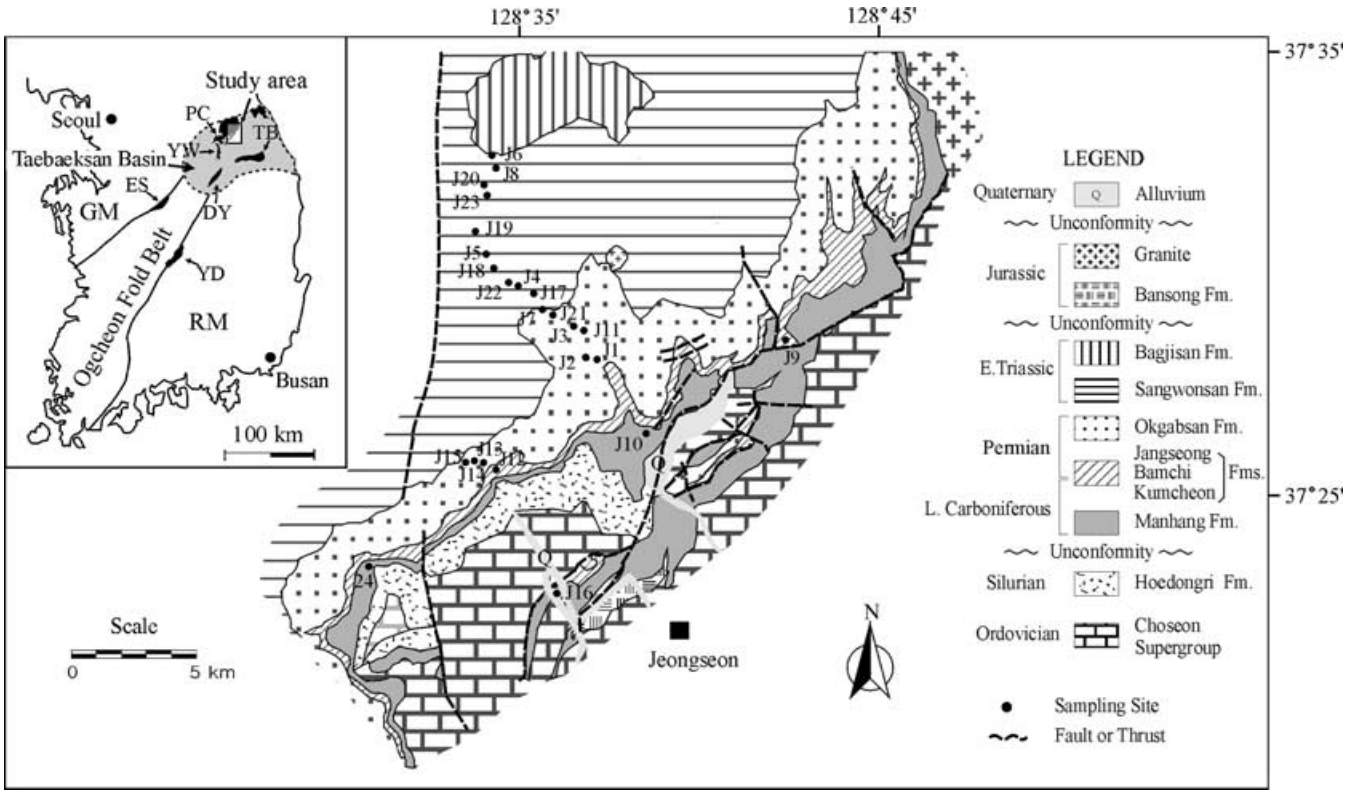


Figure 1. Geological map of the Jeongseon area (Korea Inst. Geol. and Min. Res. 1975). GM, Gyeonggi Massif; RM, Ryeongnam Massif; PC, Pyeongchang area; YW, Yeongwol area; TB, Taebaek area; DY, Danyang area; ES, Eumseong Basin; YD, Yeongdong Basin.

the southern and northern boundaries of the Ogcheon Fold Belt (e.g. the Yeongdong Basin, Doh *et al.* 1996; the Eumseong Basin, Doh *et al.* 1999). Recently, Uno (1999, 2000) reported that part of the Jeongseon area, located approximately 20 km northeast of the Yeongwol area, retained primary magnetizations, while the rest of the study area suffered complete remagnetization. This interpretation, however, can be questioned because the sampling sites revealing the primary magnetization are concentrated within a very limited area, and the reason why these sites were safe from the remagnetization is not clear. To trace the spatial distribution of remagnetized and unremagnetized areas, if any, in the Jeongseon area, a palaeomagnetic investigation of the Upper Carboniferous–Lower Triassic Pyeongan Supergroup was carried out. If the study area has been remagnetized partially or completely, the timing and process of the remagnetization can provide important insights into mineralization and geochemical processes in the study area.

2 GEOLOGICAL SETTING

In the northeastern part (the Taebaeksan Basin) of the Ogcheon Fold Belt, the Upper Carboniferous–Lower Triassic Pyeongan Supergroup is well exposed in some areas (e.g. the Taebaek, Yeongwol, Danyang, Kangneung, Pyeongchang and Jeongseon areas). The Pyeongan Supergroup in the Jeongseon area, located in the northern part of the Taebaeksan Basin, is divided into seven formations: the Manhang, Kumcheon, Bamchi, Jangseong, Okgabsan, Sangwonsan and Bagjisan formations in ascending order (Korea Inst. Geol. and Min. Res. 1975). The Manhang Formation, the basal unit of the Supergroup, consists of greenish grey to dark grey shale, sandstone and three to four beds of white to light greyish pink limestone. Carboniferous conodont fossils were found in the limestone beds of the Manhang Formation (Park 1997). The average thickness of the formation is approximately 250 m. It generally overlies the Middle Ordovician Choseon Supergroup disconformably, but it covers the Early Silurian Hoedongri Formation in the southwestern part of the study area (Fig. 1). The Kumcheon Formation occurs to the northwest of the Manhang Formation, which has an average thickness of approximately 60 m and consists mainly of grey to dark grey shale, sandstone and some lenticular grey limestone beds. The limestone beds produce the Moscovian fusulinid. The Bamchi Formation, including a Sakmarian fusulinid-bearing limestone bed (Kim 1984), was found at Hoedong-ri in the southwestern part of the study area. The thickness of the formation is approximately 30 m. The Jangseong Formation consists of alternations of dark grey to black sandstone and shale, yielding some Permian plant fossils (Lee 1999). It includes some coal beds and its thickness is approximately 120 m. The Okgabsan Formation is composed of thick light grey to dark grey sandstone interbedded with thin dark grey to greenish grey shale beds. The thickness of this formation is approximately 450 m. The Sangwonsan Formation consists of sandstone with intercalations of thin shale beds and is approximately 2 km thick. The Bagjisan Formation, the uppermost unit of the Supergroup in this area, consists of fine to medium red sandstone and red shale, and its estimated thickness is approximately 500 m (Lee 1988).

The dip directions of most formations in the study area are towards the west or northwest, and are controlled by a NE–SW-trending open fold (the Bagjisan Syncline) activated in the Middle Triassic during the Songrim Disturbance (Cluzel *et al.* 1991). The southeastern margin of the study area is deformed by N–S- or NW–SE-trending en-echelon faults and folds in Middle–Late Triassic times (Cluzel 1991) (Fig. 1). These structures were modified by later

Jurassic deformation during the Daebo Orogeny, which also caused brittle deformation (i.e. faults and thrusts) along the Sinian direction (northnortheast direction) (Lee 1988). The Bulguksa Disturbance initiated extensive igneous activity and formed many hydrothermal vein deposits and skarn deposits in the Taebaeksan Basin during the Late Cretaceous–Early Tertiary (Park *et al.* 1988).

3 EXPERIMENTAL METHODS

The samples were cored with a gasoline-powered portable rock drill and oriented with a Brunton compass in the field. A total of 302 samples from 24 sites were collected: three sites from the Manhang Formation, two sites from the Jangseong Formation, nine sites from the Okgabsan Formation and 10 sites from the Sangwonsan Formation (Fig. 1).

In the laboratory, samples were trimmed into 22 mm long cylinders. All specimens were stored in mu-metal-shielded boxes to prevent them from acquiring viscous remanence, caused by external magnetic fields such as the Earth's field. After the natural remanent magnetization (NRM) of specimens was measured using a 2G cryogenic magnetometer (model 760R) and a Molspin spinner magnetometer, the specimens were demagnetized by stepwise alternating field (AF) and thermal demagnetizations. The ASC Scientific thermal demagnetizer (model TD-48) and the Molspin AC demagnetizer were used for the thermal and AF demagnetizations, respectively. To prevent the specimens from acquiring systematic parasitic remagnetization during thermal demagnetization, they were turned over at each step. Thermal demagnetization was performed at 100, 200, 300, 350, 400, 450, 500 °C and at every 20 °C from 520 to 700 °C. To detect any chemical alteration of magnetic carriers upon heating, magnetic susceptibility was measured at each stage of thermal treatment using a Bartington magnetic susceptibility meter (model MS2). AF demagnetization was performed at field strengths of 5–30 mT with 5 mT intervals and 40–90 mT with 10 mT intervals. The palaeomagnetic data from all specimens were projected on the orthogonal vector diagram (Zijderveld 1967) and the characteristic remanent magnetization (ChRM) direction of each specimen was determined using principal-component analysis (PCA) with the anchored line fit method (Kirschvink 1980) from at least three or more points.

Isothermal remanent magnetization (IRM) acquisition experiments for representative specimens were performed using an ASC Scientific impulse magnetizer (model IM-10) and hysteresis parameters were measured with a Molspin vibrating sample magnetometer (model VSM Nuvo). The stepwise thermal demagnetization of three-axes composite IRM up to 700 °C was also conducted to distinguish both coercivity and unblocking temperature behaviours (Lowrie 1990). The IRM was imparted successively at 2.5, 0.6 and 0.1 T along the Z-, Y- and X-axes, respectively, to selected specimens, followed by thermal demagnetization. Mineralogical and chemical examinations were made by the optical microscope observations and by X-ray diffraction (XRD) on clay fractions in selected samples. To confirm the characteristic size, shape, paragenesis and composition of magnetic carriers, electron microscope observations were performed for selected samples.

4 PALAEOMAGNETIC RESULTS

Among the total of 24 sites from the study area, five sites are from red beds and 19 sites are from the greenish siltstone (six sites), dark grey sandstone (six sites) and black shale (seven sites). The number of sites in red beds is much less than other lithologies because available outcrops are very limited in the study area. The intensity of the NRM, with predominantly northerly positive directions, ranges from 0.02 to 133.91 mA m^{-1} and only approximately one-sixth is above 10 mA m^{-1}.

The demagnetization behaviour is characterized by rock type and is independent of each formation. The demagnetization of pilot samples of red beds shows that the thermal demagnetization method isolates the characteristic component more effectively, while the AF demagnetization, even at the highest field strength of 90 mT, is unable to remove the remanent magnetization successfully (Fig. 2a). The red beds reveal two distinctive types of thermal demagnetization behaviours. One shows simple decay of a northnortheasterly and moderately steep down component in the 620–680 °C range, above which the direction starts to be scattered because of weak intensity (Fig. 2b). The other behaviour shows two components of remanence. The low-temperature component with southwesterly declinations and steep inclinations, a viscous remanent magnetization (VRM) component, can be removed in the initial demagnetization stage of temperature at or below 300 °C and the converging component of northnortheasterly and moderate down directions appears up to 620–680 °C (Fig. 2c). For greenish siltstone, both AF and thermal demagnetization methods are effective in isolating the ChRM direction (Figs 2d and e). After removal of a northwesterly and moderate down component at demagnetization steps below 30 mT or 350 °C, the characteristic component of northerly declination and moderate to steep inclination is isolated in the 30–50 mT or 400–580 °C range. Samples from five sites (J11, J12, J13, J14 and J19) have very weak NRM intensity (below 0.1 mA m^{-1}) and the ChRM cannot be isolated. In the case of black shales, ChRMs are mainly isolated in the 300–400 °C range (Fig. 2f).

Two sites, showing anomalously westerly and steep down directions (J3: $D/I = 299.5^\circ/71.5^\circ$; J17: $D/I = 298.9^\circ/78.4^\circ$) with relatively large α_{95} values in geographic coordinates, are excluded from the calculation of the mean direction (Fig. 3a). All the mean directions for the Upper Carboniferous, Permian and Lower Triassic strata show that the directions in geographic coordinates are more clustered than those in stratigraphic coordinates, yielding a negative regional fold test at the 95 per cent confidence level (Table 1). The mean direction of the ChRMs recorded in magnetite is $D/I = 355.7^\circ/62.9^\circ$ ($k = 55.5$, $\alpha_{95} = 10.4^\circ$) in geographic coordinates and $D/I = 327.7^\circ/39.5^\circ$ ($k = 10.4$, $\alpha_{95} = 24.8^\circ$) in stratigraphic coordinates. The mean direction of the ChRMs carried by haematite is $D/I = 345.9^\circ/63.3^\circ$ ($k = 38.7$, $\alpha_{95} = 12.5^\circ$) in geographic coordinates and $D/I = 332.1^\circ/47.4^\circ$ ($k = 6.3$, $\alpha_{95} = 33.1^\circ$) in stratigraphic coordinates. The mean directions of magnetite and haematite components are statistically indistinguishable and reveal a negative regional fold test at the 95 per cent confidence level, indicating that the ChRMs carried by the two magnetic mineral phases were acquired at the same geological time span. Therefore, treating the haematite component and magnetite component separately is apparently meaningless. The mean direction of the ChRMs calculated from 17 site-mean directions is $D/I = 354.9^\circ/61.5^\circ$ ($k = 58.0$, $\alpha_{95} = 4.7^\circ$) in geographic coordinates and $D/I = 345.3^\circ/38.4^\circ$ ($k = 4.8$, $\alpha_{95} = 18.2^\circ$) in stratigraphic coordinates. The site-mean directions in stratigraphic coordinates are more dispersed than those in geographic coordinates (Fig. 3). Stepwise untilting test shows that the maximum clustering of ChRM directions occurs at 0 per cent untilting (Fig. 4). The parameter estimating fold test (Watson & Enkin 1993), yielding the maximum k value at 1.6 per cent untilting, agrees fairly well with the results of the stepwise untilting test. These results show that the ChRMs were

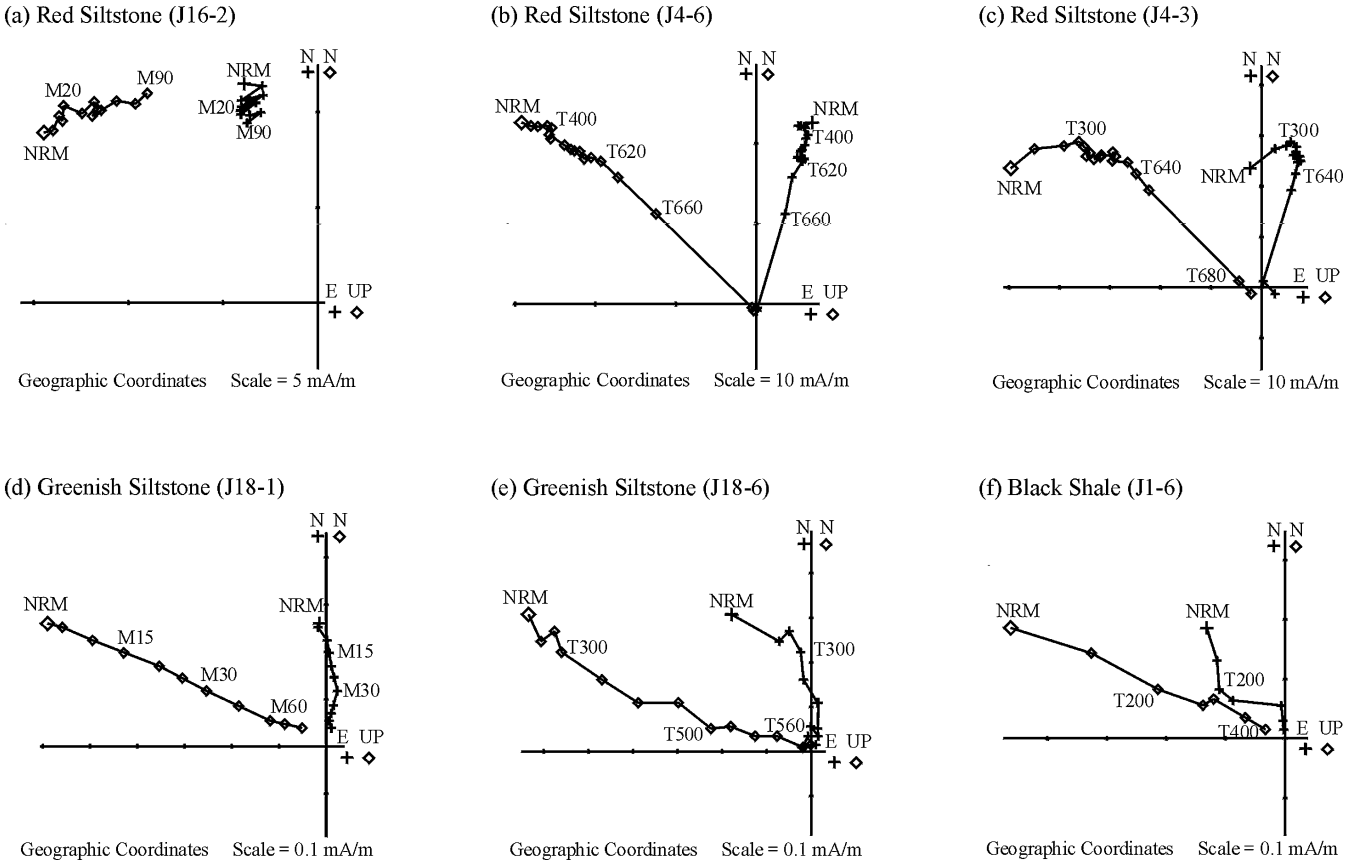


Figure 2. Representative orthogonal projections of thermal and alternating field demagnetization results of samples from red beds (a–c), greenish siltstone (d and e) and black shale (f) in the study area. Demagnetizing steps are also shown (e.g. M20: AF demagnetization at field of 20 mT; T400: thermal demagnetization at 400 °C).

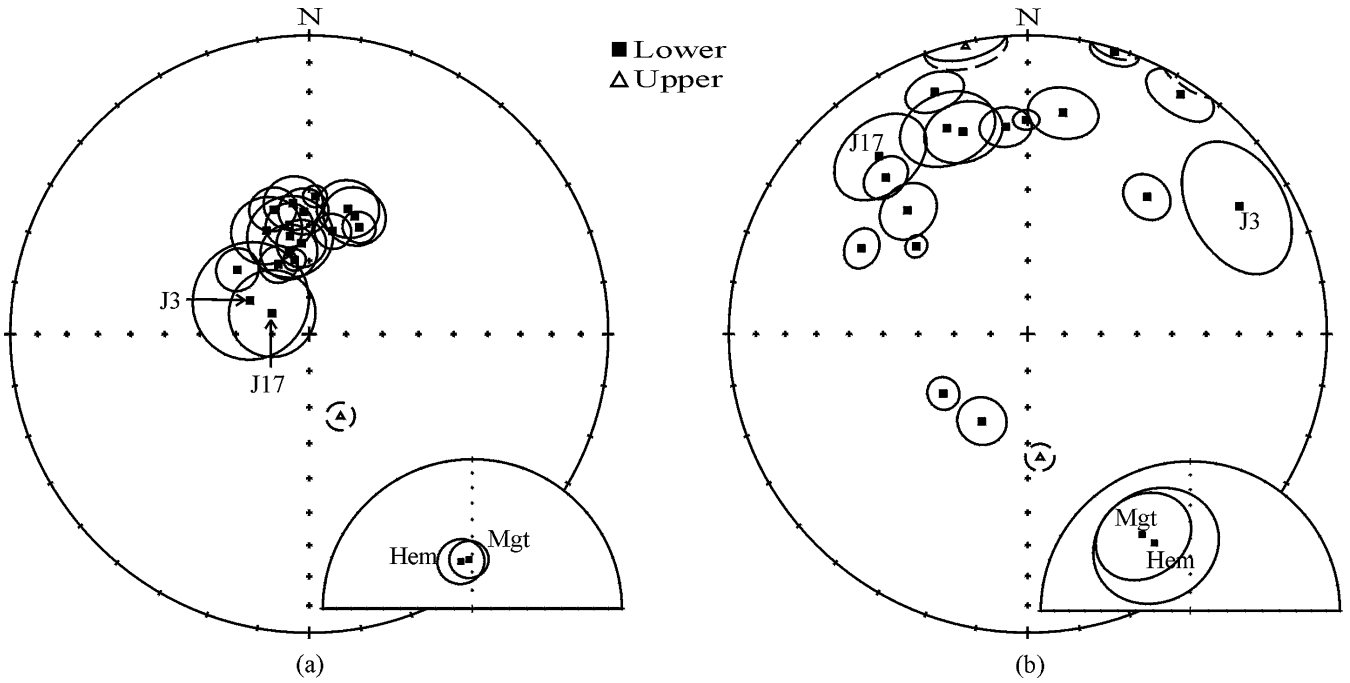


Figure 3. Equal-area projections of site-mean directions with 95 per cent confidence circles in (a) geographic and (b) stratigraphic coordinates. The mean directions of haematite (Hem) and magnetite (Mgt) components are also shown on the small equal-area projections for comparison.

Table 1. Palaeomagnetic results from the Pyeongan Supergroup in the Jeongseon area.

Formation (locality)	Site	n_u/n_m	Bedding		D_g (deg)	I_g (deg)	D_s (deg)	I_s (deg)	k	α_{95} (deg)	VGP		dp (deg)	dm (deg)
			Strike (deg)	Dip (deg)							Long. (°E)	Lat. (°N)		
							Early Triassic							
Sangwonsan (128.57°E/37.50°N)														
	J6	6/14	259	30	337.6	59.2	342.3	29.6	51.7	9.4	90.2	68.2	13.6	16.0
	J8	10/13	270	29	349.6	59.6	353.9	30.9	62.5	6.2	61.6	81.4	7.0	9.3
	J20	10/10	260	32	21.1	55.2	9.1	25.9	35.6	8.2	218.2	73.0	8.3	11.7
	J23	14/14	263	23	2.4	51.9	359.6	29.1	150.0	3.3	286.4	84.6	2.9	4.2
	J5	11/11	184	40	336.1	69.2	297.3	37.5	81.0	5.1	88.9	67.4	7.4	8.7
	J18	14/14	186	28	348.8	69.4	308.2	50.8	187.3	2.9	105.6	72.6	4.2	5.0
	J22	17/17	173	34	352.9	53.6	315.8	41.8	23.0	7.6	10.6	83.3	7.4	10.6
	J4	14/14	129	59	25.1	57.6	234.8	61.7	82.3	4.4	208.7	70.2	4.7	6.5
	J17*	8/12	236	56	298.9	78.4	320.2	23.5	23.0	11.8	100.5	45.1	21.0	22.3
	J7	16/20	102	50	355.0	65.1	207.7	63.0	32.6	6.6	109.3	79.7	8.6	10.7
	Mean	9/10			358.0	61.1			58.0	6.8	115.7	83.2	$K = 29.2$	
							328.0	50.2	5.3	24.8			$A_{95} = 9.7$	
							Permian							
Okgabsan (128.60°E/37.45°N)														
	J21	9/17	324	75	346.2	66.8	32.5	5.4	45.5	7.6	93.2	74.5	10.4	12.6
	J3*	7/11	343	84	299.5	71.5	58.9	18.4	15.3	15.9	84.1	46.3	24.4	27.9
	J15	6/11	227	72	17.2	53.8	347.8	−1.7	67.2	8.2	225.1	75.7	8.0	11.5
	J2	9/14	291	60	12.9	61.1	17.1	1.3	112.7	4.9	189.1	79.0	5.8	7.5
	J1	14/14	358	50	344.0	54.3	41.1	39.0	35.0	6.8	32.0	76.8	6.2	8.9
Jangseong (128.68°E/37.46°N)														
	J9	10/12	227	48	357.5	56.3	339.0	14.3	55.5	6.5	22.7	87.9	6.8	9.4
	Mean	5/6			0.1	59.2			71.2	9.1	127.1	87.1	$K = 37.5$	
							10.4	12.5	7.1	30.8			$A_{95} = 12.7$	
							Late Carboniferous							
Manhang (128.59°E/37.39°N)														
	J10	6/9	238	37	348.9	62.7	338.6	26.9	35.5	11.4	80.6	79.3	14.0	17.9
	J24	11/12	297	13	159.2	−65.9	174.3	−55.8	120.8	4.2	260.6	−71.4	5.1	6.2
	J16	15/20	234	34	311.8	63.6	317.8	30.0	43.9	5.8	66.6	53.5	7.3	9.2
	Mean	3/3			333.4	64.9			89.5	13.1	73.0	68.2	$K = 35.7$	
							334.6	38.4	15.0	33.0			$A_{95} = 7.8$	
Normal mean		16/18			355.7	61.2			56.6	4.9				
Reversed mean		1/1			159.2	−65.9			120.8	3.8				
Mean		17/19			354.9	61.5			58.0	4.7	97.3	82.4	$K = 28.2$	
							345.3	38.4	4.8	18.2			$A_{95} = 6.8$	

n_u/n_m, number of samples used in average/measured; D_g and I_g, declination and inclination in geographic coordinates; D_s and I_s, declination and inclination in stratigraphic coordinates; k, Fisherian precision parameter; α_{95}, radius of cone of 95 per cent confidence; VGP, virtual geomagnetic pole; dp, the semi-axis of the confidence ellipse along the great-circle path from site to pole; dm, the semi-axis of the confidence ellipse perpendicular to that great-circle path; K, the best-estimate of the precision parameter k for the observed distribution of site-mean VGPs; A_{95}, the radius of the 95 per cent confidence circle about the calculated mean pole.

*, Site mean omitted in the calculation of the formation mean direction (see the text for an explanation).

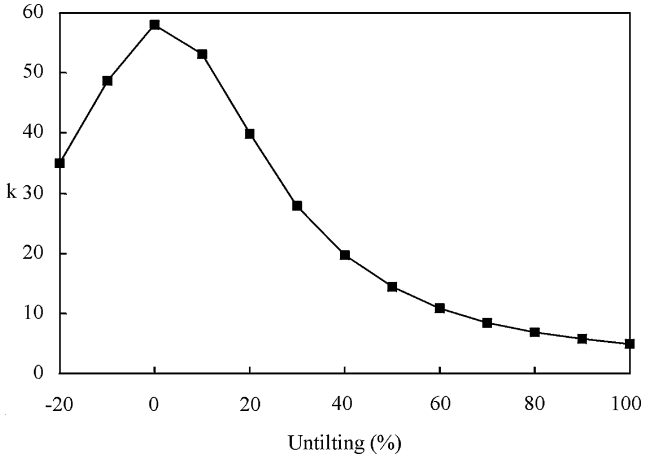


Figure 4. Syntilting test plotting Fisher's precision parameter (k) versus per cent untilting.

acquired after tilting of the strata. Among 17 site-mean directions, 16 are normally magnetized and one is reversely magnetized (Fig. 3; Table 1). The normal component ($D/I = 355.7°/61.2°$, $k = 56.6$, $\alpha_{95} = 4.9°$) and reversed component ($D/I = 159.2°/-65.9°$, $k = 120.8$, $\alpha_{95} = 3.8°$) in geographic coordinates are nearly antipodal at the 95 per cent confidence level. It therefore indicates that the remagnetization was acquired over a long enough period of time to include both normal and reversed polarity intervals.

5 ROCK MAGNETIC RESULTS

IRM acquisition experiments were performed to distinguish magnetic carriers in the samples (Fig. 5). Ferrimagnetic minerals (e.g. magnetite) are usually saturated in a field of less than 300 mT, while canted antiferromagnetic minerals (e.g. haematite and goethite) require more than 2 T to be saturated (Butler 1992). Although monoclinic pyrrhotite is a ferrimagnetic mineral, single-domain (SD) or pseudo-single-domain (PSD) pyrrhotite generally shows a

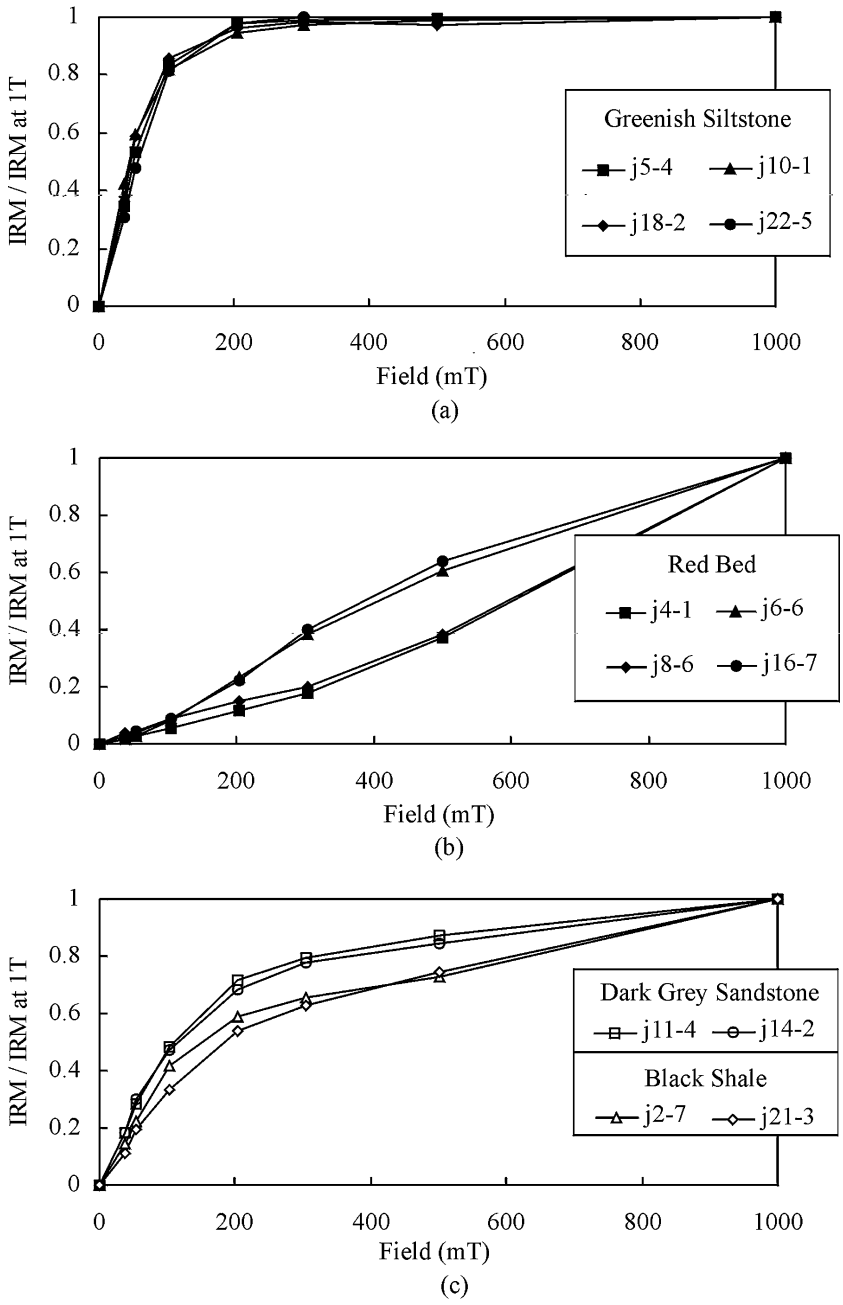


Figure 5. IRM acquisition curves of selected samples from the study area.

continuous increase in IRM intensity at relatively high field (0.3–2.7 T) (Rochette *et al.* 2001). Samples of this study can be grouped based on IRM acquisition behaviour. First, all samples from greenish siltstone show a steep increase in IRM intensity up to a field of 100 mT and then saturated at 200 mT (Fig. 5a). This behaviour suggests that ferrimagnetic material makes an important contribution to the IRM acquisition. Secondly, samples from red beds show a continuous increase in IRM intensity with increasing field up to 1 T (Fig. 5b), revealing that canted antiferromagnetic material is a dominant magnetic phase contributing to the IRM acquisition. The third group, samples from the dark grey sandstone and black shale, shows that the IRM is acquired rapidly up to 200 mT and then increases gradually without saturation (Fig. 5c). This behaviour is an indication of either the mixture of ferrimagnetic and canted antiferromagnetic materials or the presence of SD and/or PSD pyrrhotite.

The stepwise thermal demagnetization of three-axes composite IRM was conducted for representative samples (Fig. 6). Samples of greenish siltstone reveal that the soft- to medium-coercivities with maximum unblocking temperatures of 580 °C are attributed to magnetite (Fig. 6a). Fig. 6(b) shows that haematite is the major magnetic phase in the red beds, which is conformed by the soft- to hard-coercivities with unblocking temperature up to 680 °C (Fig. 6b). In samples of dark grey sandstone, magnetite with relatively low unblocking temperatures is the dominant magnetic phase (Fig. 6c). Most samples of black shales show the removal of the hard-coercivity component at approximately 100 °C and that of the soft- to medium-coercivity components at 350 °C during the stepwise

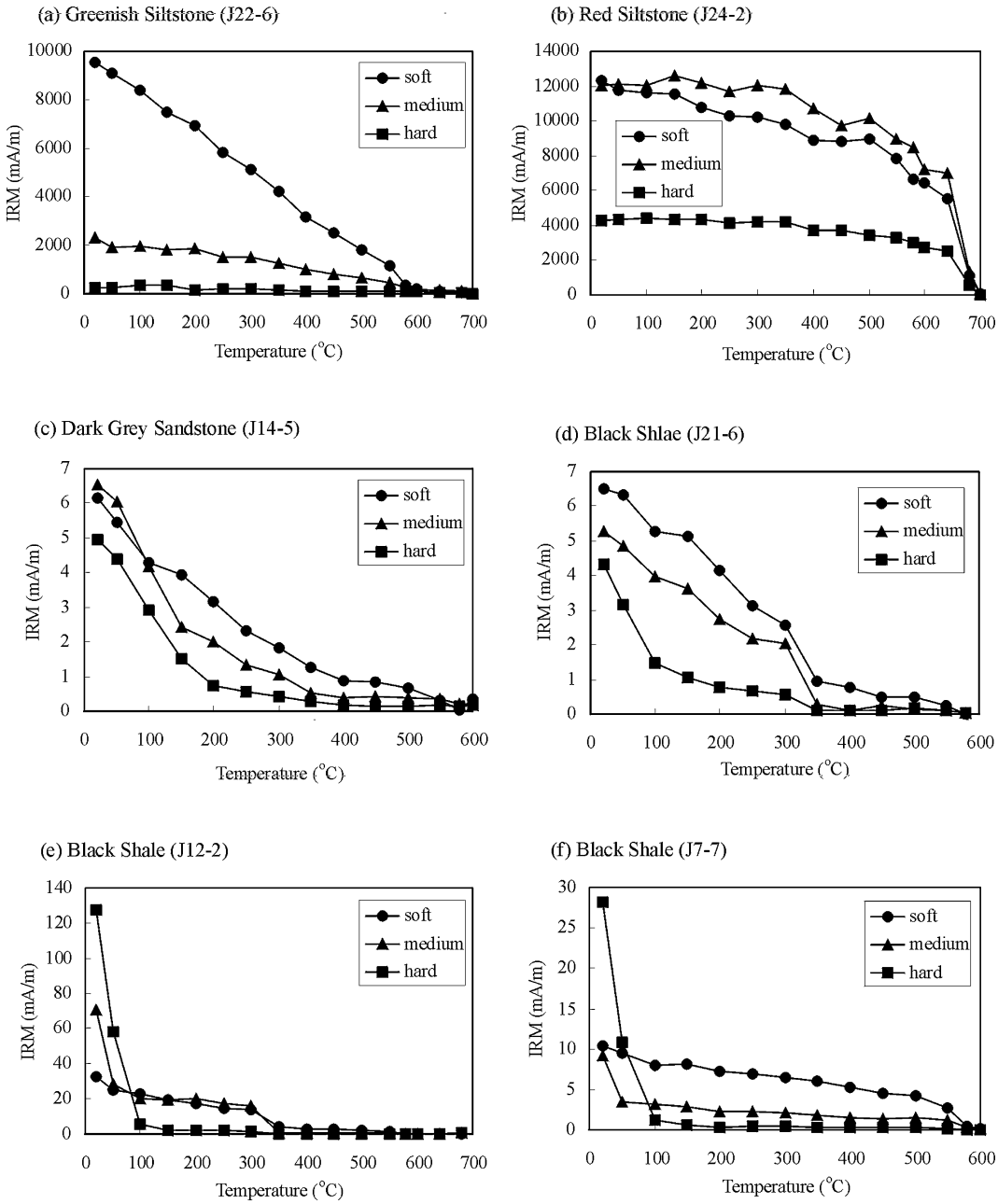


Figure 6. Thermal demagnetization results of three-axes composite IRM for the representative samples.

thermal demagnetization, indicating the presence of goethite and pyrrhotite, respectively (Figs 6d and e). The coexistence of goethite and magnetite is less frequently observed in black shales (Fig. 6f).

Hysteresis parameters (saturation magnetization, M_s; saturation remanence, M_{rs}; coercive force, H_c; coercivity of remanence, H_{cr}) for the selected samples from the study area are listed in Table 2. Minerals with corundum structure, such as haematite, have H_{cr} above 200 mT, while those with spinel structure, such as magnetite, have H_{cr} below 50 mT (Thompson *et al.* 1980). Red beds have H_{cr} above 290.9 mT, indicating that haematite is a dominant magnetic phase, while greenish siltstone and dark grey sandstone samples have H_{cr} below 81.4 mT, indicating that magnetite is a main magnetic carrier with little influence of haematite or goethite. Relatively high H_{cr} (144.8–223.4 mT) and low Curie temperature in black shales are indicative of goethite. These results are in accordance with the grouping based on the IRM acquisition behaviour (Fig. 5) and the thermal decay of three-axes composite IRM (Fig. 6). The ratios of H_{cr}/H_c and M_{rs}/M_s (i.e. the Day plot) are used to diagnose the domain state of magnetic minerals (Day *et al.* 1977). Most samples of greenish siltstone belong to the PSD region, while dark grey sandstone samples are plotted in the MD region (Fig. 7). Although the Day plot is used to determine the domain state of magnetite-bearing samples, the results from red beds and black shales are also shown in Fig. 7.

The results of the modified Lowrie–Fuller test (Lowrie & Fuller 1971; Johnson *et al.* 1975) for greenish siltstone displayed that anhysteretic remanent magnetization (ARM) is more stable than IRM against AF demagnetization below 50 mT, yielding an SD or PSD test result (Figs 8a and b). Fig. 8(c) showed the inflected shape of decay curve, which is evidence for a considerable contribution from

Table 2. Hysteresis parameters of selected samples.

No.	H_{cr} (mT)	H_c (mT)	H_{cr}/H_c	M_{rs} (μAm^2)	M_s (μAm^2)	M_{rs}/M_s	Rock type
J4-1	538.18	54.63	9.85	7.54	45.75	0.16	Red siltstone
J6-6	327.91	27.60	11.88	13.53	81.85	0.17	Red siltstone
J8-6	420.19	219.67	1.91	39.40	155.00	0.25	Red siltstone
J16-7	290.93	76.50	3.80	1.83	24.63	0.07	Red siltstone
J5-4	49.15	14.62	3.36	9.32	90.30	0.10	Greenish siltstone
J10-1	79.30	60.08	1.32	14.70	32.02	0.46	Greenish siltstone
J18-2	43.55	14.66	2.97	1.14	8.49	0.13	Greenish siltstone
J22-5	49.23	4.18	11.78	2.07	82.78	0.03	Greenish siltstone
J11-4	81.39	7.36	11.06	0.26	50.64	0.01	Dark grey sandstone
J14-2	80.06	12.19	6.57	0.88	81.92	0.01	Dark grey sandstone
J2-7	223.42	5.10	43.81	0.34	60.47	0.01	Black shale
J21-3	144.84	3.21	45.12	0.02	8.41	0.00	Black shale

H_{cr}, coercivity of remanence; H_c, coercive force; M_{rs}, saturation remanence; M_s, saturation magnetization.

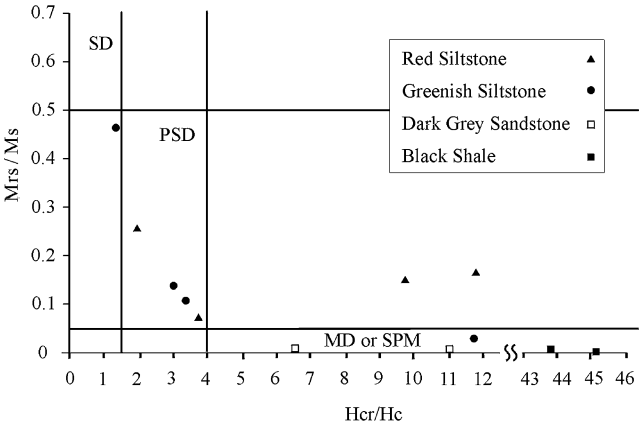


Figure 7. Hysteresis properties and domain state for selected samples from the study area (after Day *et al.* 1977).

SD and PSD grains (Dunlop 1983). Because both ARM and IRM imparted in haematite or other minerals of high H_{cr} are rarely removed by AF demagnetization, this test is inadequate for red beds and black shales.

6 ELECTRON MICROSCOPE OBSERVATIONS AND XRD ANALYSES

Electron microscope observations were carried out for representative samples of greenish siltstone and red siltstone from the study area. To identify the magnetic carriers and their relation with the adjacent grains or matrix, compositional analyses using energy dispersive spectroscopic (EDS) system for X-ray analysis were performed. In order to distinguish the magnetic phase between magnetite and haematite, the morphological characteristics and the iron–oxygen ratio determined by optical and electron microscopic observations, the results of the rock magnetic experiments, and thermal demagnetization behaviour of samples were collectively used. In samples of greenish siltstone, most magnetic carriers are too small to be detected at the scanning magnification of 2000× and iron oxides large enough to be detected are rarely observed. This suggests that the domain state may be SD or PSD, agreeing well with the results of the modified Lowrie–Fuller test and the Day plot. Detrital iron oxide grains were known to show eroded features and some relic phases such as titanomagnetite, titanohaematite or rutile (Suk *et al.* 1990). In a greenish siltstone sample, a non-spherical pure magnetite, filling the pore space between quartz and clay minerals, does not show any detrital characteristics (Fig. 9a). This feature implies that magnetite in greenish siltstone might be formed authigenically after deposition.

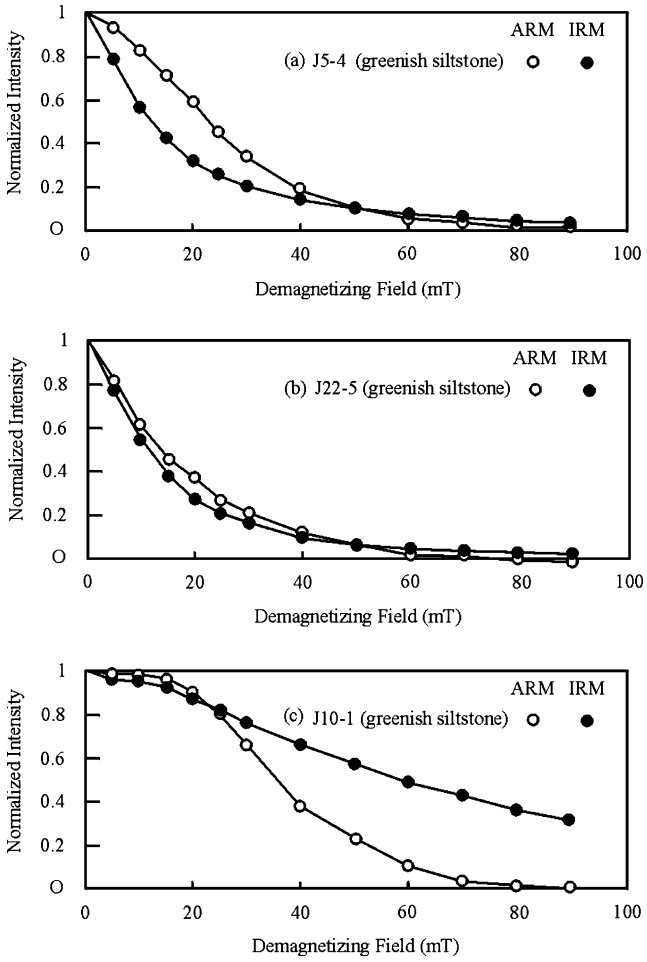


Figure 8. The modified Lowrie–Fuller test (Lowrie & Fuller 1971; Johnson *et al.* 1975), showing SD/PSD behaviour in greenish siltstone.

Haematite grains are the most frequently observed iron oxides in samples of red siltstone and are commonly accompanied by quartz, feldspar and clay minerals. Although rarely encountered, detrital Fe-Ti oxide grains showing severe corrosion features of an

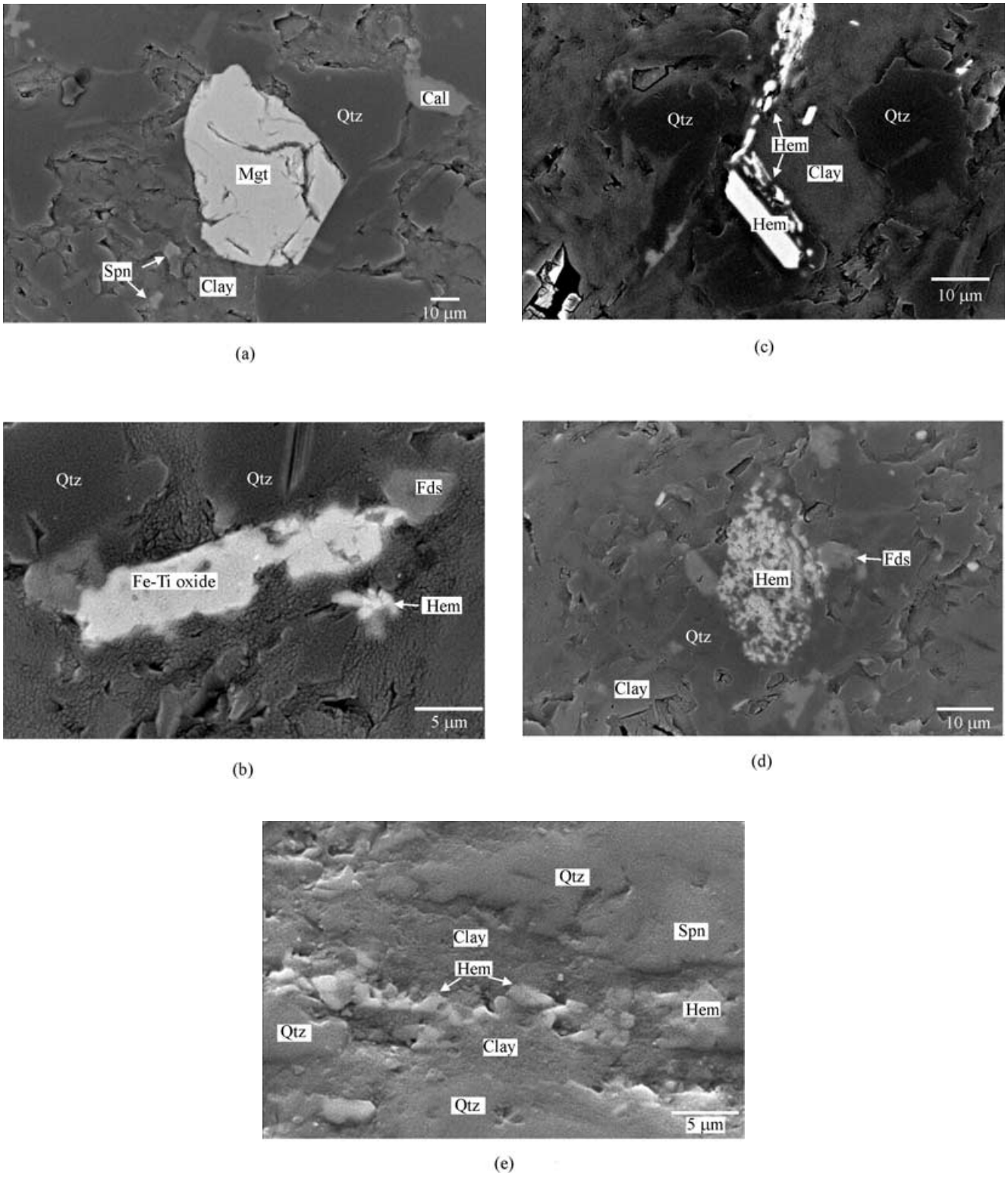


Figure 9. Scanning electron microscopy photographs of samples from greenish siltstone (a) and red siltstone (b–e). Mgt, magnetite; Qtz, quartz; Spn, sphene; Cal, calcite. Hem, haematite; Chl, chlorite; Fds, feldspar. (a) Back-scattered electron image (BEI) of non-spherical magnetite filling pore space between adjacent quartz and clays. (b) BEI of Ti-bearing iron oxide and authigenic haematite without Ti. (c) BEI of a euhedral haematite grain and pigmentary haematite grains scattered along grain boundaries. (d) BEI of authigenic haematite as aggregates of individual grains. (e) Secondary electron image (SEI) of haematite grains embedded within the clay cements.

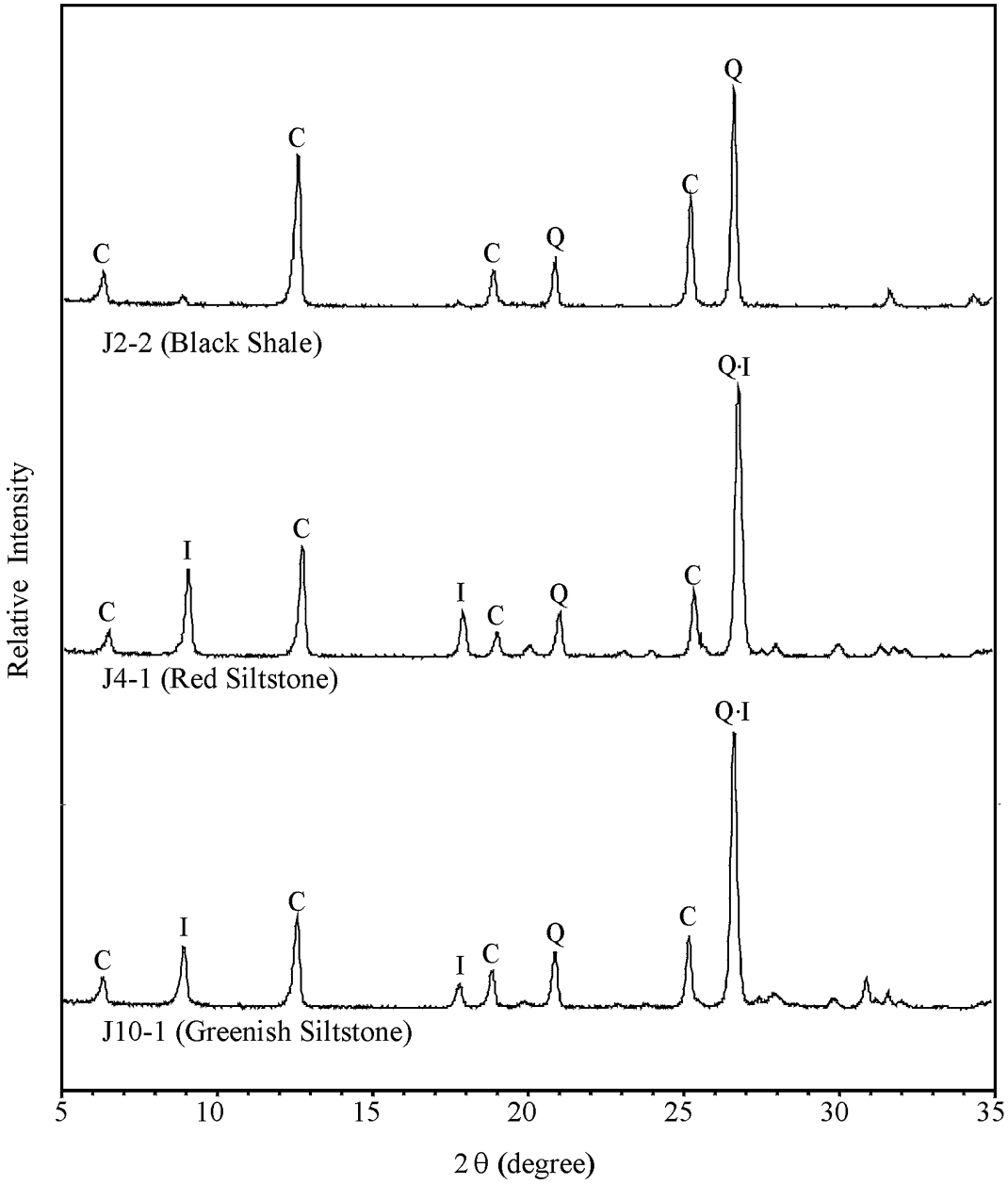


Figure 10. Representative XRD patterns of the air-dried powder specimens of <2 μm fractions of selected samples from the study area. C, chlorite; I, illite; Q, quartz.

originally large elongated grain are observed (in the centre of Fig. 9b). Pure iron oxide grains, presumably authigenic haematite, are also observed next to the Fe-Ti oxides. Euhedral haematite and small pigmentary haematite grains distributed along grain boundaries are observed in a red siltstone sample (Fig. 9c). Moreover, many aggregates of submicron-sized haematite grains distributed within an area of approximately 20 μm diameter are observed in quartz grains (Fig. 9d). In Fig. 9(e), haematite grains are embedded in Fe-bearing clay minerals, indicating that they were formed by precipitation from Fe-bearing fluids, for which Fe is enriched during the processes of alteration of Fe-bearing minerals to clays (Walker *et al.* 1981). These features imply that most haematite grains in red siltstone generally have authigenic characteristics.

The XRD analyses were performed for the air-dried powder specimens of <2 μm fractions of selected samples to determine the clay mineralogy. XRD patterns show the distinct peaks of illite, chlorite and quartz (Fig. 10), indicating that illite and chlorite are dominant phases of clay, while kaolinite and smectite are minor or absent. This result implies that rocks in the study area were altered significantly through illitization and chloritization. It was reported that conversion of smectitic minerals to illite could result from burial diagenesis at depths of 2–3.5 km (Chamley 1989). Iron ions, which are major ingredients for magnetite authigenesis, can be released during illitization (Hirt *et al.* 1993; Katz *et al.* 2000). Therefore, it can be assumed that the magnetite authigenesis is related to the burial diagenesis in the study area. The alteration of the rocks, however, can be caused by not only burial, but also interactions between fluids and rocks. Potassium can be supplied from the decomposition of K-feldspars associated with the illitization (Chamley 1989) or orogenic-type fluids (McCabe *et al.* 1989). The results of the

electron microscope observations and the XRD analyses suggest that the major magnetic carriers (e.g. magnetite and haematite) are secondary products formed after sedimentation. Magnetite authigenesis can occur in relation to burial diagenesis or fluid-mediated alteration under reduced conditions, whereas secondary haematite may form under oxidized conditions. Hence, both haematite and magnetite hardly ever form simultaneously. The existence of both haematite and magnetite will be discussed in Section 7.2.

7 DISCUSSION AND CONCLUSIONS

7.1 Remagnetization of the Pyeongan Supergroup in the Jeongseon area

In this study we show that the Pyeongan Supergroup in the Jeongseon area has been undoubtedly remagnetized. Contrary to our results, Uno (2000) reported the palaeomagnetic direction (D/I = 347.4°/23.8°, $\alpha_{95} = 5.1°$) obtained from the samples of greenish sandstone in a small part of the study area as a primary remanent magnetization. In the study by Uno (2000), the primary component was isolated from four sites that are within the same region as sites J8, J20 and J23 of this study. Fig. 11 shows the results of Uno (2000) compared with those from the corresponding sites (J8, J20 and J23). The tilt-corrected directions of Uno (2000) and this study do not seem to be different and the directions before tilt correction are also similar except for one site (NF1) of Uno (2000). The mean direction in geographic coordinates calculated from three sites excluding the site NF1 of Uno (2000) is not statistically different at the 5 per cent significance level from the Late Cretaceous and Tertiary palaeomagnetic directions in Korea. The mean direction of three sites (J8, J20 and J23) of this study is $D/I = 4.9°/56.2°$ ($k = 72.4$, $\alpha_{95} = 14.6°$) in geographic coordinates and $D/I = 1.0°/28.8°$ ($k = 126.4$, $\alpha_{95} = 11.0°$) in stratigraphic coordinates. Although the directions in geographic coordinates are more dispersed than those in stratigraphic coordinates, it is interpreted that the Pyeongan Supergroup in the study area has been remagnetized, because we cannot find any differences to distinguish the directions of these three sites from those of the other sites in this study (Fig. 3). The site-mean direction showing shallow inclination in geographic coordinates (e.g. NF1 of Uno 2000) does not appear in this study. Additional arguments contrary to the result of Uno (2000) can be raised. Uno (2000) isolated the high-temperature component in the 670–690 °C range from the greenish siltstone. In contrast, all the samples of greenish siltstone in our study show magnetite behaviour during thermal and AF demagnetizations (Figs 2d and f) and in the rock magnetic studies (Figs 5 and 6). Moreover, during the thermal demagnetization of red beds, the ChRMs were isolated in the 620–680 °C range, above which the direction started to be scattered because of weak intensity (Fig. 2b).

Doh & Piper (1994) reported the Early Triassic palaeomagnetic direction ($D/I = 331.1°/19.4°$, $\alpha_{95} = 18.2°$) from the red beds in the Taebaek area (the Baekunsan Syncline). Also, the Lower Triassic Tonggo Formation in the Danyang area revealed the mean direction ($D/I = 121.4°/-26.4°$, $\alpha_{95} = 7.2°$) as the primary remanent magnetization (Doh *et al.* 1998). In contrast to these areas, the Yeongwol area was remagnetized by the acquisition of a CRM during the Tertiary (Doh *et al.* 1997). Comparison of the results of this study with those of the previous palaeomagnetic studies for the adjacent regions suggests that the complete remagnetization occurred in the northwestern part (e.g. the Jeongseon and Yeongwol areas) of the Taebaeksan Basin, whereas the primary remanent magnetization has been retained in the southeastern part (e.g. the Taebaek and Danyang areas) of the basin. This result implies that each part of the basin has experienced different geological events since the deposition of the strata. Cheong (1982) reported the significant difference of palaeontologic facies in the Pyeongan Supergroup between the southeastern and northwestern parts of the Taebaeksan Basin. The Sakmarian fusulinids occurred only in the northwestern part, indicating a hiatus in the southeastern part during the Sakmarian. Cheong (1982) also pointed out that the structural trends in the northwestern region are in the NS direction, while those in the southeastern region are NE–SW and ENE–WSW. Choi *et al.* (1992) argued that the effect of structural deformation on the northwestern part was much stronger than that on the southeastern part since the Late Permian. These reported palaeontologic data and structural characteristics of the Taebaeksan Basin are consistent with the results of this study in the sense that the pervasive remagnetization in the northwestern region was apparently connected to the severe structural deformations.

7.2 Mechanism of remagnetization in the Jeongseon area

Several possible mechanisms can cause remagnetization. First, surface weathering should be excluded because all magnetic carriers (magnetite, haematite, goethite and pyrrhotite) recorded extremely well-clustered directions ($\alpha_{95} = 4.7°$), which are hard to acquire by weathering processes. More importantly, magnetic grains found by electron microscope observations cannot be produced by weathering processes. Secondly, a thermoviscous remanent magnetization (TVRM) origin for the remagnetization can be proposed. The estimated maximum palaeotemperatures of 340 °C, as determined by electrum-sphalerite geothermometry and fluid inclusion studies of

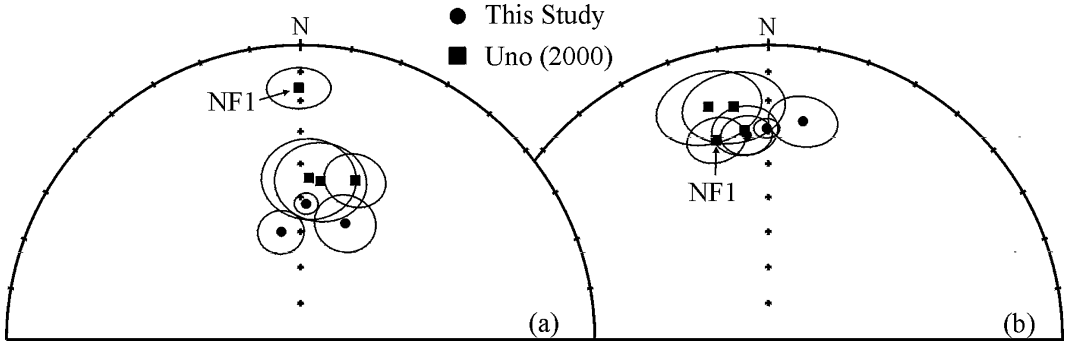


Figure 11. Equal-area projections of site-mean directions from four sites (NF1, NF2, NF3 and NF4) of Uno (2000) and from the corresponding sites (J8, J20 and J23) of this study in (a) geographic and (b) stratigraphic coordinates.

several hydrothermal veins in the Jeongseon area (Park *et al.* 1988; Park & Park 1990; Park & Hwang 1992), is significantly lower than temperatures required to have activated the remagnetization for haematite with an unblocking temperature of 620–680 °C, as determined by the blocking temperature–relaxation time curves for haematite (Pullaiah *et al.* 1975). Although the characteristic component that resides in goethite and pyrrhotite could be a TVRM based on the relation between the unblocking temperatures of these magnetic phases and the maximum palaeotemperature (340 °C) of the study area, a ChRM in pyrrhotite is less likely to be a TVRM since the relaxation temperatures for pyrrhotite do not vary with time as much as for magnetite (Dunlop *et al.* 2000). The palaeotemperature of 340 °C might also produce a TVRM in magnetite with an unblocking temperature below 500 °C, as determined by the blocking temperature–relaxation time curves for magnetite (Pullaiah *et al.* 1975). However, the stable single-ChRM component residing in magnetite was isolated in the 400–580 °C range, indicating that both magnetizations with unblocking temperatures below and above 500 °C are the same component, which cannot be explained by a TVRM model. The directions of magnetite and pyrrhotite components with maximum unblocking temperatures of 580 and 320 °C, respectively, are consistent with the direction of the haematite component with maximum unblocking temperatures of 680 °C; this result implies that all the components were acquired by a single process at almost the same time. Based on rock magnetic studies and electron microscope observations, the major magnetic carriers are SD–PSD haematite and magnetite of secondary (authigenic) origin. Hence, the CRM is most likely to be the major mechanism of remagnetization in the study area.

Traditionally, many have reported CRM to be connected to the tectonically driven external fluid migration over long distances (Oliver 1986) or the *in situ* pore waters related to tectonic pressure (Elmore *et al.* 1993). Recently, Katz *et al.* (1998, 2000) proposed that the burial diagenesis of clays rather than the alteration by orogenic-type fluids caused the widespread chemical remagnetization in SE France. On the other hand, Henry *et al.* (2001) refuted the results of Katz *et al.* (1998) and proposed that the CRM in SE France is related to topographically driven fluid flow and burial diagenesis. It is known that intense hot fluid circulations, associated with modifications of porosity and permeability of sediments, induced large authigenesis of illite and reworking of pre-existing clay minerals in eastern France (Tournier *et al.* 1999). Although the results of our XRD study, revealing abundant secondary illite and chlorite without any smectite, agree with those of Katz *et al.* (2000), the burial diagenesis model is hard to accept because the Pyeongan Supergroup in the Jeongseon area was not be buried to depths (e.g. more than 2 km) deep enough to cause the burial diagenesis and remagnetization during the Late Cretaceous to Tertiary (Lee 1988). Therefore, it is interpreted that a fluid-mediated process is most likely to be responsible for acquisition of the CRM.

The nature of fluid related to the acquisition of the CRM is still problematic. There are many skarn and hydrothermal vein deposits, especially the meso-epithermal-type deposits, near the study area. Most meso-epithermal-type deposits in the Taebaeksan Basin are genetically associated with Late Cretaceous granitic activity (Park *et al.* 1988). Previous geochemical and isotopic data showed that all of the Late Cretaceous deposits exploited or explored in Korea formed from isotopically evolved meteoric water at depths of mostly <1 km (e.g. Shelton *et al.* 1990; So *et al.* 1990). On the other hand, the important role of primary magmatic water in the formation of the hydrothermal system has been documented as a common feature in mesothermal-type gold–silver deposits in Korea (So & Yun 1997). Hur & Park (2000) reported the hydrogen and oxygen isotope compositions of hydrothermal fluids from the Shinjeongseon (SJ) and Sinch (SC) mines near the study area. The ore-fluids $\delta^{18}O_{water}$ (2.7 to 3.9 ‰, SJ; 5.0 to 5.8 ‰, SC) and δD_{water} (−59 to −58 ‰, SJ; −67 to −56 ‰, SC) values indicate mixed magmatic (dominant)–meteoric fluids. Although it has not been proved that these mineralizing fluids circulated the host rocks pervasively, we speculate that the CRMs in the study area are related

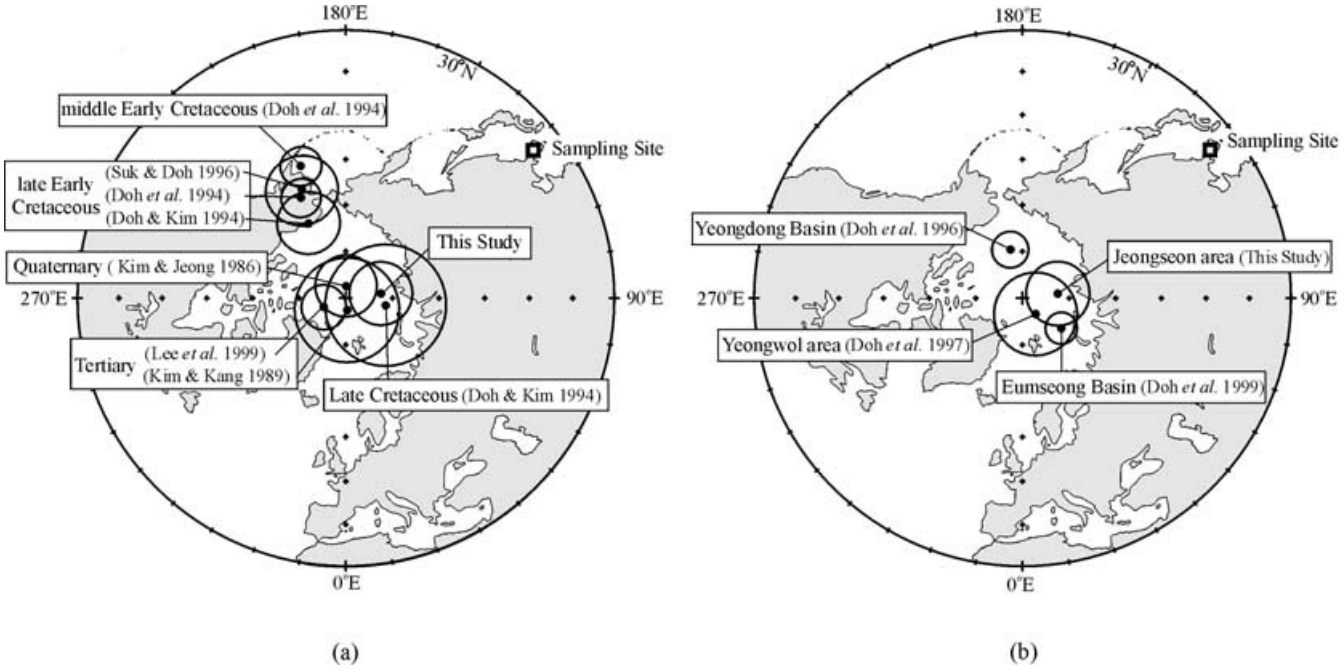


Figure 12. Palaeomagnetic pole calculated from the remagnetized component of the Pyeongan Supergroup in the Jeongseon area compared with (a) middle Early Cretaceous to Quaternary poles obtained in Korea and (b) those from remagnetized sedimentary rocks in the Yeongwol, Eumseong and Yeongdong basins.

to the mixed magmatic–meteoric fluids. The regional distribution of remagnetized and unremagnetized areas within the Taebaeksan Basin, as stated in the previous section, can be explained by the connection of the CRMs with the extent of structural deformation. There are many thrusts in the more intensively deformed northwestern region, while there are a limited number of thrusts in the southeastern region (Ree *et al.* 2001). It can be interpreted that the remagnetizing fluids penetrated more easily and pervasively into the northwestern region by utilizing the already well-developed thrust system.

Rock magnetic studies suggest that the CRMs reside in diverse magnetic phases depending on rock types: haematite in red beds, magnetite in greenish siltstones and goethite/pyrrhotite in black shales (Figs 5 and 6). It can be interpreted that the authigenesis of these magnetic phases were mainly controlled by the lithology and the oxidation state of the host rocks rather than the chemical composition of fluids because all CRMs in each magnetic phase were acquired at almost the same time. The mixed magmatic–meteoric fluid propagated in the graphite-containing black shales formed goethite and pyrrhotite under reduced conditions, while the fluid in red beds under oxidized conditions caused haematite precipitation. The fluid in greenish siltstones of less oxidizing condition relative to the red beds might form magnetite.

7.3 Timing of remagnetization in the Jeongseon area and its implications

The palaeomagnetic pole calculated from the remagnetized component of this study is at 97.3°E, 82.4°N (A_{95} = 6.8°). This palaeomagnetic pole position is close to those of the Late Cretaceous, Tertiary and Quaternary poles in Korea (Fig. 12a). Thus, it can be interpreted that the remagnetization occurred during the Late Cretaceous–Quaternary. Because any remarkable geological events, such as orogeny or igneous activity that might cause the

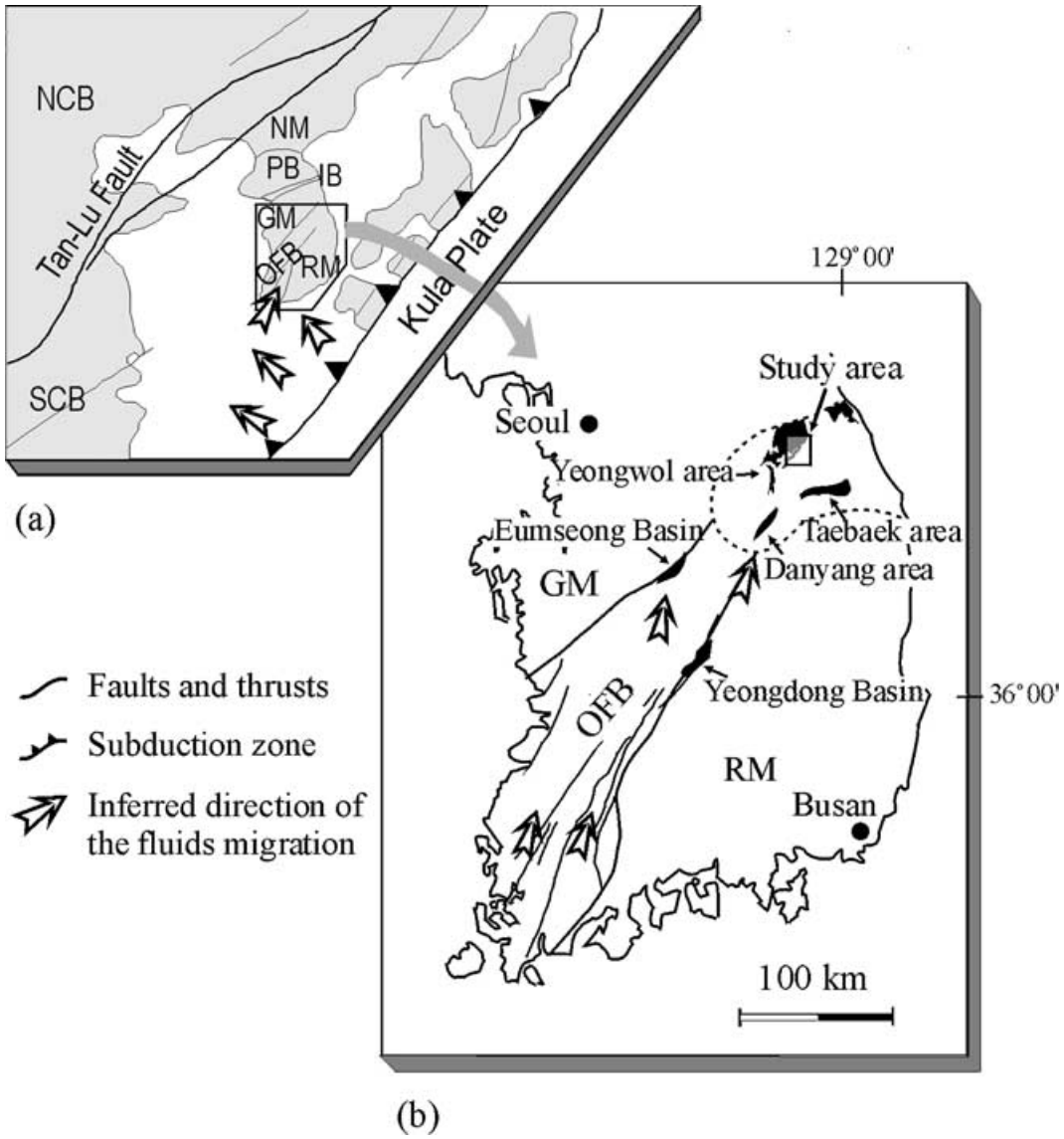


Figure 13. (a) Tectonic Map of the Korean Peninsula and adjacent major blocks. (b) Simplified map of Korea showing the remagnetized areas cited in this study and inferred directions of the fluids migration in and along the Ogcheon Fold Belt during the Late Cretaceous to Early Tertiary. NCB, North China Block; SCB, South China Block; NM, Nangrim Massif; PB, Pyeongnam Block; IB, Imjingang Belt; GM, Gyeonggi Massif; OFB, Ogcheon Fold Belt; RM, Ryeongnam Massif.

remagnetization, are not recognized during the Quaternary, the remagnetization in the study area could have occurred during the Late Cretaceous–Tertiary. A number of hydrothermal vein deposits in the northeastern subzone of the Ogcheon Fold Belt formed between 94 and 73 Ma (Park *et al.* 1988), indicating that the fluid-mediated process related to the Bulguksa igneous activity occurred during the Late Cretaceous. Some K-Ar age dating data (51.97 ± 1.8 Ma of phlogopite in Wondong mine, Park *et al.* 1988; 56–67 Ma of rhyolite and quartz porphyry in Taebaeksan region, Shin & Jin 1995) imply that the fluid-mediated process would have remained active until the Early Tertiary. Hence, the CRM of the study area is interpreted as having been acquired due to the formation of the authigenic minerals during the Late Cretaceous–Early Tertiary under the influence of the hydrothermal fluids that might be triggered by the Late Cretaceous Bulguksa igneous activity. Similar aspects of the chemical remagnetization are also observed in several areas such as the Yeongwol area (Doh *et al.* 1997), Eumseong Basin (Doh *et al.* 1999) and Yeongdong Basin (Doh *et al.* 1996). The palaeomagnetic pole position calculated from the remagnetized component of the Yeongdong Basin is distinguishable from the Late Cretaceous and Tertiary pole positions and those of the other areas (Fig. 12b). This difference implies that fluid-mediated chemical remagnetization in the Yeongdong Basin and the other areas occurred at significantly different times.

Okada (2000) reported that the Late Cretaceous arc magmatism of East Asia (the Bulguksa igneous activity in the Korean Peninsula) was related to the oblique subduction of Kula/Pacific plates under the Eurasian Plate. It is also reported that the basinal fluid events were triggered by the Kula–Eurasia convergence and migrated through the fault system within the Ogcheon Fold Belt during the Cretaceous period (Sillitoe 1977; Kong *et al.* 1997) (Fig. 13a). The major faults within the Ogcheon Fold Belt were activated initially during the Triassic (the Songrim Disturbance) and mainly in the Late Jurassic (the Daebo Orogeny) and later reactivated and formed several Cretaceous basins (e.g. the Yeongdong, Eumseong, Gongju and Poongam basins). It is also known that cratonic fault zones typically have long histories of recurrent activity that may span hundreds of millions of years, whereas fault zones in actively deforming mountain belts have a relatively short-lived history (Marshak & Paulsen 1996; Ramsey & Onasch 1999). Therefore, it is interpreted that the initiation of fluid migration related to the magmatism responsible for the remagnetization events triggered by the plate subduction propagated northward or northeastward until the Early Tertiary in and along the boundaries of the Ogcheon Fold Belt (Fig. 13b).

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